

WPIS: From In-the-Wild Web Images to Physics-Aware 3D Scene Graphs for Physical Reasoning

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Abstract

Intelligent agents operating in biomedical laboratories need physics-aware understanding beyond simple geometry or semantics. In-the-wild web images capture authentic lab interactions, yet methods designed for instrumented 3D capture and reconstruction struggle to turn this resource into functional knowledge. Consequently, current autonomous laboratory systems lack a queryable representation of affordances for precise object and liquid handling tasks. We pose a web-native question: how to convert single-view, uncalibrated web images into a structured, physics-aware scene representation. We introduce WPIS (Web- and Physics-Informed Scene understanding), a pipeline that compiles Physics-aware 3D Scene Graphs (P-3DSGs) from web imagery by fusing open-vocabulary instance/mask cues with relative geometry, augmenting nodes with real-valued liquid states and fine-grained hand-object interaction (HOI) subgraphs, and attaching concise natural-language functional

relations—without intrinsics, multi-view, or CAD priors. We release WebLab-3DSG, a 1,000-scene knowledge base pairing each RGB image with its P-3DSG JSON, relative depth, and a single-image point-cloud proxy. In an expert study, grounding an LLM in a P-3DSG improves answer quality by ~40% over a strong VLM that reasons directly from RGB, with the largest gains on feasibility, HOI alignment, and safety constraints. WPIS offers a reproducible path to physics-aware reasoning from in-the-wild web imagery and a practical substrate for decision-making in autonomous lab settings.

CCS Concepts

• Computing methodologies → Knowledge representation and reasoning.

Keywords

Physics-aware 3D scene graphs, Web image-to-graph construction, Hand-object interaction, Laboratory automation

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1 Introduction

Biomedical laboratories are undergoing a rapid transition from fixed, hard-coded workcells to *autonomous, AI-guided laboratories (ABLs)* that combine computer vision, robotics, and process automation [3, 5, 17, 18, 33, 45, 55, 57]. In these settings, liquid handling (e.g., dispensing, aspiration, mixing) is both ubiquitous and safety-critical: it demands high accuracy and strict protocol adherence (sterility, contamination control), and its feasibility depends on the *physical state* of the manipulated container and contents [13, 22, 34]. Despite advances across perception and automation, what is often missing is a *physics-aware, queryable knowledge base* that tells an agent *how* to safely manipulate liquid-filled containers and interactive lab tools. Current ABL stacks rarely provide such a structured source of *affordance* knowledge. Meanwhile, the web contains a vast, untapped repository of real laboratory interactions (SOP photos, vendor pages, tutorials) that depict humans operating liquid-filled containers and tools; however, these images are unstructured and not compiled into a representation usable for reasoning. This work therefore tackles a web-native question: turning uncalibrated, single-view images on the web into a structured, physics-aware knowledge base for safe laboratory manipulation.

3D Scene Graphs (3DSGs) offer a structured substrate for entities, attributes, and relations, and have supported navigation and task planning [1, 2, 32, 39]. Yet most pipelines assume high-fidelity 3D inputs (posed RGB-D, laser scans) [7, 10, 50] and favor qualitative spatial predicates (“on”, “next to”). Recent progress moves toward *open-vocabulary* and *functional* 3DSGs [26, 62] and benchmarks how LLMs interface with graphs [58]. However, these advances are typically built on instrumented capture with known intrinsics, depth, multi-view, or CAD priors—assumptions that do not hold for in-the-wild web imagery. This leaves open a core web question: *How can we construct a functional, physics-aware scene graph from single-view, uncalibrated web images?*

Functionality in biolabs is *physics-conditioned*: safe tilt, aspirate/dispense feasibility, or rack stability depend on real-valued physical state variables (e.g., liquid fill, liquid centroid, support polygon) that generic SG vocabularies do not capture. Meanwhile, fine-grained HOI—grasp posture, contact region, approach direction—directly governs sterility and safety. Public HOI resources [4, 8, 46] quantify pose and contact for everyday objects, but they are not tied to *liquid-aware* affordances or compiled at web scale for lab operations.

We embrace this web-first setting and build from *single-view, uncalibrated* RGB. Traditional metric reconstruction is ill-posed, but modern components, monocular relative-depth estimation [40, 41], open-vocabulary detection/segmentation [25, 31, 49], and VLMs [27, 29], provide reliable *relative geometry* and semantics at web scale. Our key idea is to compile these heterogeneous signals into a P-3DSG whose geometry is relative (scale-ambiguous) yet sufficient for reasoning about safety and feasibility.

We present **WPIS**, a web-to-graph pipeline that compiles in-the-wild laboratory images into a **P-3DSG** expressly for modeling *physics-aware affordances*, illustrated in Figure 1. WPIS (i) fuses open-vocabulary instance/mask cues with monocular relative depth to derive *quantitative spatial vectors* between object centroids and surfaces; (ii) augments nodes with *real-valued liquid states* (fill

level, liquid centroid) and *HOI subgraphs* (hand posture, contact); and (iii) attaches lightweight *physics constraints* (capacity bounds, support/containment, spill stability). From these components, the P-3DSG *instantiates state-conditioned affordances*, e.g., stability, pourability, aspirate/dispense safety, that can be queried for downstream decisions. The graph encodes uncertainty and prioritizes relational consistency over absolute metric accuracy. Unlike prior functional 3DSGs [62], our representation explicitly binds real-valued content states to the affordances that matter in ABLs [51].

We compile the resulting P-3DSGs into **WebLab-3DSG**, a domain-specific, queryable knowledge base. The current release contains **1,000** samples from distinct laboratory scenes, and each sample includes an RGB image, a P-3DSG in JSON format encoding the three-level hierarchy, relations, and state-conditioned affordances, an estimated depth image, and a generated point cloud derived from the RGB image. Rather than training end-to-end policies, we use WebLab-3DSG to ground an online agent’s LLM reasoning with structured, physically plausible priors distilled from web data. This echoes evidence that planners and LLMs benefit from structured SG priors [1, 39, 58] while introducing the missing layers of liquid-aware and HOI-aware semantics [26, 62].

Our contributions can be summarized as: (1) A web-to-graph methodology (**WPIS**) that turns single-view web images into *hierarchical, physics-aware* P-3DSGs with relative spatial vectors, real-valued liquid states, and fine-grained HOI. (2) **WebLab-3DSG**, a public knowledge base for laboratory manipulation. (3) Evidence that *LLM reasoning grounded in P-3DSG* improves safety/feasibility judgments over raw VLM prompting, complementing open and functional 3DSG literature [26, 62].

2 Related Work

Our work lies at the intersection of 3D scene graphs, affordance understanding, perception for autonomous biomedical laboratories, hand-object interaction, and web-scale knowledge compilation with LLM grounding. We emphasize how our P-3DSGs complement and extend these lines.

2.1 3D Scene Graphs: From Spatial to Functional

Classic 3D scene graphs unify objects, attributes and relations in a layered graph and have been built predominantly from posed RGB-D or laser scans [2, 7, 10, 50]. Online incremental pipelines, such as SceneGraphFusion [53] and Hydra [23, 24], construct hierarchical graphs from sensor streams, supporting robotics navigation and planning [1, 39]. More recently, open-vocabulary 3DSGs leverage foundation models to generalize beyond closed taxonomies [19, 26] and to scale to large indoor spaces.

However, conventional 3DSGs mostly encode *spatial* relations (e.g., *on*, *next to*) between object nodes. Two concurrent threads push towards richer semantics. First, hierarchical room-region-object graphs associate semantics with spatial context [23, 24, 50]. Second, *functional* 3DSGs introduce *interactive elements* (handles, switches) and explicit *functional edges*, enabling action-centric reasoning [62]. TB-HSU further shows that affordances vary with spatial granularity and should be modeled across layers (objects/regions/rooms) [56]. Our work is complementary: rather than relying on posed

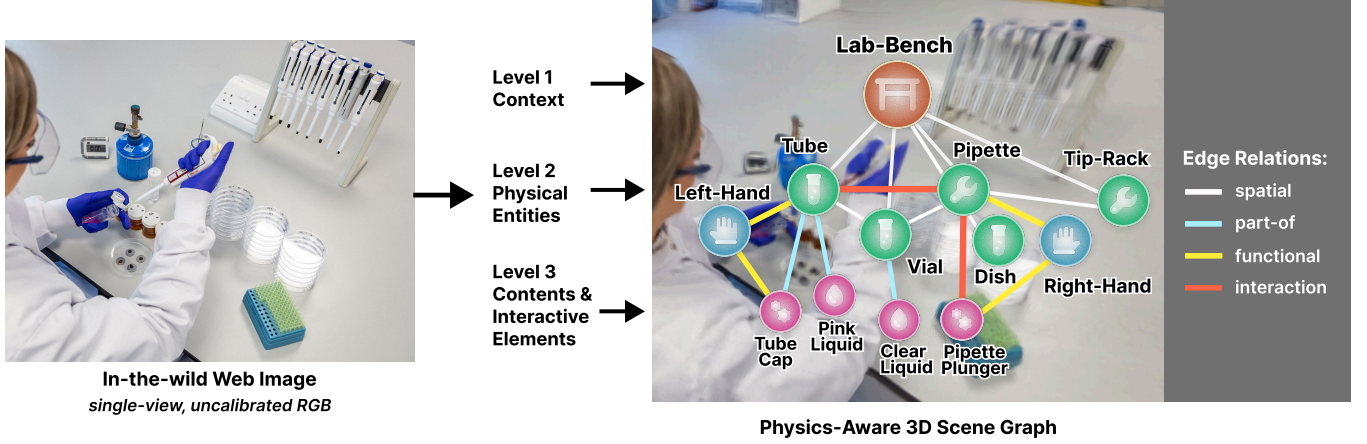


Figure 1: WPIS overview. From a single uncalibrated web image, we build a P-3DSG. Right: the three-level graph (context, physical entities, contents/interactive elements) with typed edges (spatial, part-of, functional, interaction) is overlaid on a single-image point cloud reconstructed from the RGB, enabling affordance-centric reasoning.

RGB-D inputs, we synthesize relative 3D geometry and physics-aware state variables from single-view web images and encode (i) continuous liquid states, (ii) contact-aware HOI subgraphs, and (iii) quantitative spatial vectors with uncertainty. This bridges open-vocabulary SGs [19, 26] and functional graphs [62] with physics-aware semantics needed in biomedical labs.

2.2 Affordance Understanding

Affordance learning spans 2D/3D inputs and supervision granularities [47, 48]. On the 3D side, 3D AffordanceNet [12] establishes a large-scale shape-level benchmark and tasks (full-shape/partial-view/rotation-invariant). SceneFun3D [11] focuses on *fine-grained interactive elements* and motion cues in real indoor scenes, and has been adopted by functional 3DSG methods [62]. Beyond static objects, TB-HSU highlights that affordances depend on spatial context granularity [56]. The above advances, however, rarely model continuous physical states (e.g., liquid fill and center-of-mass shift) that modulate feasible actions. Our P-3DSG explicitly stores these continuous states and links them to handling constraints (e.g., tilt limits, spill risk), complementing categorical affordance vocabularies.

2.3 Perception for ABLs

Constructing a physics-aware graph for laboratory manipulation requires a stack of representations: *2D detections or masks, monocular or RGB-D depth, scene or instance point clouds, 6D object poses, and liquid states*. Domain-specific resources are emerging but scarce. Vector-LabPics targets vessel recognition and contents in chemistry scenes [15]; DTLT formalizes *dual transparency* in biomedical labs and benchmarks *liquid-level estimation* with multi-view capture [51]; LCDTC benchmarks liquid content detection in transparent containers [54]. System-level lab pipelines often rely on RGB-D to infer 6D labware poses and volumes under controlled setups [43].

By contrast, state-of-the-art 6D pose estimation typically assumes strong priors: intrinsics/extrinsics, RGB-D/multi-view, and often CAD or reference views. FoundationPose improves generalization to novel objects but still presupposes reference models

or views and benefits from RGB-D for tracking [52]. Benchmarks such as BOP require calibration and model availability [20, 21]. Transparent-object geometry further challenges RGB-D; methods employ synthetic training [42], multi-view/stereo consistency [6], challenge benchmarks on specular/transparent surfaces [38], polarization cues [44], or depth completion [28]. These pipelines improve perception under instrumentation, yet their input assumptions do not hold for in-the-wild web images.

Motivated by this gap, we embrace *single-view web RGB* and replace missing metric priors with *relative 3D* geometry (centroid-to-centroid/surface vectors; ordinal depth) derived from monocular depth and open-vocabulary segmentation, while estimating liquid-aware continuous states. The resulting P-3DSG stores uncertainty and physical consistency checks, enabling downstream reasoning.

2.4 Hand-Object Interaction

HOI datasets (ContactPose, DexYCB, GRAB) provide hand/object pose and contact supervision for everyday manipulation [4, 8, 46]. They are collected in controlled environments and seldom condition interaction strategy on *object internal states* (e.g., a full bottle requires a different grasp due to shifted COM). For lab tasks (pipetting, vial capping/uncapping), there is no public HOI-3D dataset with *liquid-aware affordances*. Our web-informed approach instead distills HOI patterns from large-scale web imagery and encodes them as HOI subgraphs linked to physics-aware states.

2.5 Web-Scale Compilation and LLM Grounding

Modern components enable extracting *relative* geometry and semantics at scale from single images: MiDaS/DPT for monocular relative depth [40, 41], models for open-vocabulary detection and segmentation [16, 25, 31, 36, 64], BLIP-2/LLaVA for multimodal attribution [27, 29, 37, 63]. ConceptGraphs [19] and task-driven Clio [35] show LLM/VLMs can interface with 3D graphs for planning. OpenFunGraph [62] demonstrates leveraging foundation models to infer functional graphs with interactive elements. Concurrently, TSG-Bench probes how LLMs understand/generate scene graphs

[58]. We contribute a domain-specific, physics-aware graph constructed from web imagery that an LLM can query for safety and feasibility judgments in biomedical manipulation.

3 Problem Formulation

3.1 P-3DSGs

Classical 3D scene graphs represent objects, attributes and relations for indoor environments [2, 50], and recent work extends them toward open-vocabulary or functional semantics [62]. For autonomous biomedical labs, we require a representation that (i) encodes *relative, camera-centric 3D geometry* suitable for single-view images, (ii) includes *real-valued physical state variables* (e.g., liquid fill in $[0, 1]$) that modulate feasible actions, and (iii) records *interaction structure* (hands and interactive elements).

A scene is a directed, attributed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{L}, \mathcal{A})$. Nodes $\mathcal{V}$ are typed entities; edges $\mathcal{E}$ are typed relations between ordered node pairs. The level map $\mathcal{L} : \mathcal{V} \rightarrow \{1, 2, 3\}$ partitions nodes into:

- **Level 1: Context** (e.g., work surface, large appliances),
- **Level 2: Physical Entities** (e.g., containers, tools, hands, supports),
- **Level 3: Contents & Interactive Elements** (e.g., liquids; caps, plungers).

Attributes $\mathcal{A}$ include real-valued states for relevant nodes and relation parameters for edges. For a pair (v_i, v_j) , spatial edges store a scale-ambiguous, camera-centric vector $\mathbf{r}_{ij} = [\Delta x, \Delta y, \Delta z]^\top$ and a contact indicator; $\mathbf{r}_{ij}$ captures *relative* layout.

3.2 Task Definition

Single-view web-to-graph. Given a single, uncalibrated RGB web image $\mathcal{I}$ and optional text $\mathcal{T}$, predict a P-3DSG $\mathcal{G}$ that includes: (i) a node set with types across the three levels; (ii) spatial edges with relative vectors $\{\mathbf{r}_{ij}\}$ and contact indicators; (iii) functional / containment / part-of / interaction edges; (iv) real-valued physical state variables (e.g., liquid fill level); (v) state-conditioned affordances.

Assumptions. No camera intrinsics/extrinsics, no depth or multi-view, and no CAD models are available at test time. Geometry is therefore relative yet sufficient for downstream physical reasoning.

4 Methodology: The WPIS Pipeline

We present **WPIS**, a web-to-graph pipeline that converts a single, uncalibrated RGB image $\mathcal{I}$ into a P-3DSG. The framework is presented in Figure 2. WPIS follows the schema in the Problem Formulation, $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{L}, \mathcal{A})$, with three levels, typed nodes/edges, and real-valued attributes. Two design principles make the graph physics-aware: (i) estimate *state variables* that govern feasibility and safety (e.g., liquid fill_level); and (ii) bind those states to *state-conditioned affordances* (stability, pourability, aspirate feasibility) and to *functional relationships* expressed both structurally and in concise natural language for LLM reasoning.

Tracks. We separate a **compilation** track (for building the WebLab-3DSG knowledge base with human verification) from an **inference** track (fully automated). Geometry is camera-centric and relative; uncertainty proxies are used for confidence-aware reasoning.

Table 1: Corpus-level statistics for WebLab-3DSG

Category	Item	Count / Rate
Nodes	WorkSurface	992
	Container	5,987
	Tool	1,127
	Hand (total)	1,372
	SupportingObject	1,463
	Substance	3,967
	InteractiveElement	2,935
	Appliance	317
Edges	spatial_relation	17,842
	interaction_relation	1,621
	functional_relation	1,308
	part_of_relation	1,087
	containment_relation	3,912
HOI types	actuating	518
	holding	403
	manipulating	362
	stabilizing	294
	pouring/tilting	211
	capping/uncapping	163
	placing/positioning	147
	opening/closing	121
	presenting/hand-off	96
	wiping/sterilizing	85
Task categories [†]	Pipetting/Transfer	732
	Tube/Rack Handling	561
	Plating/Dispense	423
	Reagent Handling	497
	Aseptic/Flaming	182
	Vortex/Mixing	263
	Storage/Cap/Seal	338
	Measurement/Weighing	119
	Labeling/Documentation	112
Liquid state	Mean fill level ($\mu \pm \sigma$)	0.52 ± 0.25
Contact	Contact ratio over spatial edges	12.1%
Safety gear	Hands wearing gloves	94.7% (rate)

[†] Multi-label taxonomy.

4.1 Stage I: Web Data Collection and Curation

Reproducible API-driven acquisition. We use Microsoft Bing APIs and log all query strings, timestamps, and result URLs.

Phase 1 (keyword seeding). Long-tail queries combine domain and context keywords (e.g., biomedical lab, cell culture, protocol, tutorial, experiment, youtube video) with supplier keywords (e.g., Corning, Thermofisher, Sigmaaldrich). We query the Image Search endpoint with a resolution filter (width>640, height>480) and manually select 100 diverse, high-quality seed images.

Phase 2 (visual-similarity expansion). For each seed, we retrieve visually similar images using Bing’s Visual Search API v7 (“similar-image insights”) and deduplicate by perceptual hashing (pHash) with a tuned Hamming threshold to retain distinct scenes.

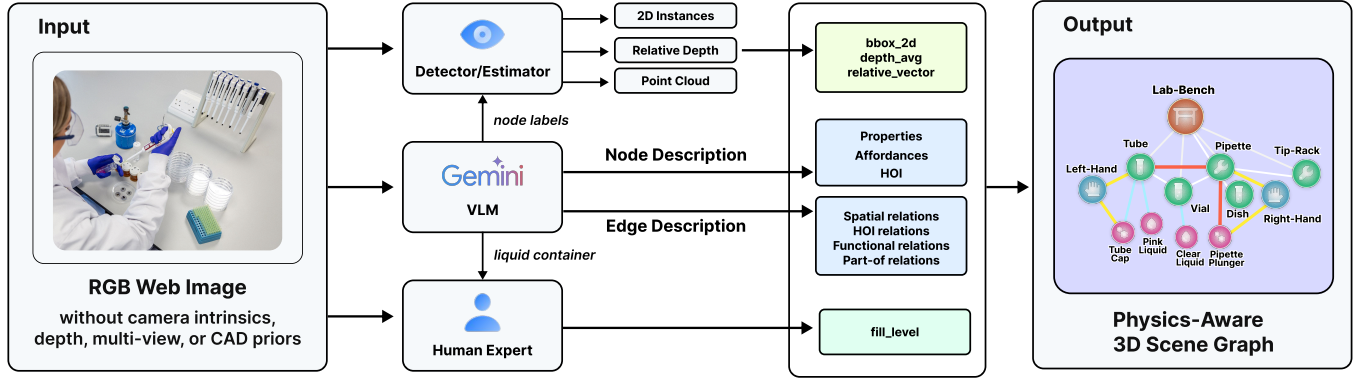


Figure 2: The WPIS framework: from a single web RGB image (no intrinsics/depth/multi-view/CAD), three branches—detector/estimator (2D instances/masks, relative depth), VLM (node/edge text), and expert verification—plus a single-image point-cloud cue collaborate to populate a P-3DSG with nodes (bbox_2d, depth_avg, fill_level, HOI) and edges (spatial, functional, part-of).

Table 2: Top entities and hand posture tokens (1,000 scenes).

Category	Item	Count
Top labels	Left-Hand	672
	Right-Hand	700
	Pipette-Plunger	612
	Lab-Bench	604
	Biosafety-Cabinet-Surface	431
	Red-Cell-Culture-Media	394
	Single-Channel-Micropipette	407
	Pipette-Tip-Box	389
	Pink-Liquid	413
Posture tokens	Clear-Liquid	378
	positioned_on_plunger	593
	standard_pipette_grip	417
	body (contact region)	403
	supporting_dish_base	214
	stabilizing_dish_edge	197
	precision_pinch_grip	223
	top (contact region)	206
	side (contact region)	193

Curation and authenticity. We keep an image only if it (i) depicts an authentic laboratory setup with workspace context, (ii) contains visible hand–labware interaction, (iii) is sufficiently sharp for detection/masks, and (iv) is not synthetic/AI-generated or staged advertising with unnatural layouts.

Ethics, licensing, and logging. We respect site ToS/robots, exclude faces/PHI, maintain source URLs/timestamps/file hashes.

Final scope and intent. After curation, the **WebLab-3DSG knowledge base** contains **1,000** unique RGB images (one per distinct scene). In Figure 3, we select nine image samples representing distinct tasks in ABLs. The scale is intentionally modest: it is sufficient to validate the feasibility of WPIS and to assess the effectiveness of P-3DSG for physics-aware reasoning from web imagery.



Figure 3: Representative WebLab-3DSG scenes spanning nine ABL task families.

4.2 Stage II: From 2D to Pseudo-3D

Task-aware open-vocabulary detection. A VLM (Gemini 2.5 Pro) proposes task-relevant categories (containers, tools, interactive elements, hands) via a few-shot prompt with a compact ontology (short disambiguations, e.g., “plunger: thumb-actuated cylinder on a pipette”). These labels condition Grounding DINO [31] to localize instances and output (bbox_2d, mask, label, s_{det}) per instance. Instances must pass a confidence threshold and low-score cases are deferred to a VLM verification pass (Stage III).

Monocular relative depth. We obtain a per-pixel relative depth map directly from $\mathcal{I}$ using a transparent-object depth estimator (MODEST), trained on ClearPose to improve robustness on transparent interfaces [9, 30]. As web images lack camera priors and metric ground truth, we use depth only as an *ordering* signal. MODEST predicts D_{raw} ; we resize to the detector/mask resolution and

Section	ID/Pair	Type	Label/Kind	Quantitative (bbox_2d, depth_avg, etc.)	Qualitative (material, state, contact)	Affordance / NL description
Node	scene_1	WorkSurface	Lab-Bench	—	—	A researcher at a lab bench is using a single-channel micropipette to transfer liquid between a vial and a test tube.
Node	appliance_1	Appliance	Bunsen-Burner	bbox: (cx=501, cy=258, w=108, h=200); depth_avg: 0.30	state: off	Provides an open flame for sterilization or heating when connected to a gas source.
Node	obj_1	Container	Centrifuge-Tube	bbox: (cx=450, cy=406, w=100, h=80); depth_avg: 0.20	material: plastic; state: open	Can be capped for mixing (e.g., vortexing) or storage. Suitable for holding small liquid volumes for reactions or centrifugation.
Node	obj_2	Container	Brown-Glass-Vial	bbox: (cx=509, cy=433, w=46, h=66); depth_avg: 0.40	material: brown glass; property: light-blocking; state: closed	Used for storing chemical reagents, especially those sensitive to light.
Node	obj_3	Tool	Single-Channel-Micropipette	bbox: (cx=602, cy=344, w=290, h=172); depth_avg: 0.18; volume_range: 100–1000 μ L	—	Designed for precise aspiration and dispensing of liquid volumes.
Node	obj_4	Hand	Left-Hand	bbox: (cx=423, cy=424, w=172, h=112); depth_avg: 0.20	wearing_gloves: true; interaction: to obj_1; type: stabilizing	palm_fingers: cradling_tube_body, thumb: resting_on_upper_body; contact_points: forefinger on body, middle_finger on body
Node	obj_5	Hand	Right-Hand	bbox: (cx=677, cy=317, w=154, h=166); depth_avg: 0.15	wearing_gloves: true; interaction: to obj_3; type: actuating	palm_fingers: supporting_pipette_body, thumb: positioned_on_plunger; applying_pressure_to_plunger
Node	obj_7	Container	Petri-Dish-Stack	bbox: (cx=847, cy=488, w=302, h=284); depth_avg: 0.45; count: 3	material: plastic	—
Node	obj_8	SupportingObject	Pipette-Tip-Rack	bbox: (cx=850, cy=680, w=176, h=192); depth_avg: 0.40	—	—
Node	content_1	Substance	Pink-Liquid	fill_level: 0.10	type: liquid; in_container: obj_2	—
Node	content_2	Substance	Clear-Liquid	fill_level: 0.20	type: liquid; in_container: obj_1	—
Node	elem_1	InteractiveElement	Pipette-Plunger	—	type: plunger	—
Node	elem_2	InteractiveElement	Tube-Cap	bbox: (cx=465, cy=391, w=43, h=34); depth_avg: 0.22	type: cap	—
Edge	obj_1 to elem_2	spatial	—	rv: [−0.122, −0.013, −0.419]	contact: false	—
Edge	obj_3 to obj_1	spatial	—	rv: [−0.144, −0.016, 0.296]	contact: false	—
Edge	obj_2 to elem_2	spatial	—	rv: [−0.08, 0.03, −0.2]	contact: true	—
Edge	obj_7 to obj_8	spatial	—	rv: [−0.015, 0.024, 0.2]	contact: true	—
Edge	obj_1 to appliance_1	spatial	—	rv: [0.014, 0.013, −0.42]	contact: false	—
Edge	obj_3 to obj_1	functional	—	—	—	The micropipette tip is inside the centrifuge tube, likely dispensing or aspirating liquid.
Edge	elem_1 to obj_3	functional	—	—	—	The plunger controls the liquid handling action of the micropipette.
Edge	obj_4 to obj_1	interaction	—	—	—	Left-Hand interacting with Centrifuge-Tube (stabilizing).
Edge	obj_5 to obj_3	interaction	—	—	—	Right-Hand interacting with Single-Channel-Micropipette (actuating).
Edge	elem_2 to obj_1	part-of	—	—	—	Tube-Cap is part of Centrifuge-Tube.
Edge	obj_1 to content_1	part-of	—	—	—	Container holds Pink-Liquid.
Edge	obj_2 to content_2	part-of	—	—	—	Container holds Clear-Liquid.

Table 3: Example P-3DSG. Quantitative columns report numeric fields (bbox_2d, depth_avg, counts, fill_level, rv); qualitative columns capture categorical cues (material, state, wearing_gloves, contact); the rightmost column lists natural-language affordances (hand posture, contact points, actuation). Edges shown are a representative subset.

normalize per image to $[0, 1]$:

$$D = \frac{D_{\text{raw}} - \min(D_{\text{raw}})}{\max(D_{\text{raw}}) - \min(D_{\text{raw}})}.$$

Let $\text{mask}(v)$ be the instance mask for node v (same resolution as D). We assign a relative instance depth by averaging within the mask:

$$\text{depth_avg}(v) = \text{mean}_{p \in \text{mask}(v)} D(p).$$

Averaging reduces local noise near transparent boundaries and yields a stable scalar per instance for downstream relative 3D reasoning. When confidence-aware reasoning is used, we keep a dispersion term as a soft uncertainty proxy,

$$\sigma_D(v) = \text{std}_{p \in \text{mask}(v)} D(p),$$

otherwise it is omitted (or set to zero).

Point cloud generation. Independently of depth, we obtain a camera-centric point cloud $\tilde{X}$ *directly from the RGB* using a single-image 3D scene generation model (Marble 01-mini). Such methods (e.g., WonderJourney, WonderWorld) reconstruct an interactive 3D proxy from a single view, prioritizing plausible *layout* over exact geometry [14, 60, 61]. We upload one RGB image to Marble (<http://marble.worldlabs.ai/>), fix `seed=0` and model `Marble 01-mini`, and export the point cloud. Although shapes are not metrically accurate and no instance-level meshes are produced, the cloud encodes stable relative spatial arrangement sufficient for non-contact relations and physics-aware affordances.

4.3 Stage III: P-3DSG Construction

Given the evidence from Stages I–II, we assemble the P-3DSG $\mathcal{G}$ following the fixed three-level schema and a strict attribute-provenance policy. This stage makes physics-awareness explicit.

Node induction and attribute provenance. We instantiate nodes top-down and attach attributes from three sources: detection or estimation, VLM-generated language, and human annotation. *Level 1 (Context).* We create WorkSurface and Appliance nodes whenever the detector returns consistent bbox_2d and mask. For each node we compute depth_avg by averaging the per-image-normalized relative depth within the mask, and obtain a concise label and description from Gemini 2.5 Pro using the crop. *Level 2 (Physical Entities).* We add Container, Tool, Hand, and SupportingObject nodes from the detector; every node carries bbox_2d, mask, depth_avg, and a VLM-generated label/description. Hands also receive wearing_gloves and a structured interaction_states list that, for each target object, records interaction_type (e.g., stabilizing, grasp, actuate), a short posture_description (e.g., thumb on plunger, cradling_tube_body, and contact_points as the finger-region pairs). *Level 3 (Contents & Interactive Elements).* We create liquid Substance and InteractiveElement (e.g., cap, lid, plunger) nodes with bbox_2d, mask, depth_avg, label, and description. The only real-valued state attached here is fill_level for Substance nodes; due to the difficulty of quantifying volume from a single RGB view, fill_level $\in [0, 1]$ is provided by trained annotators and later used to gate affordances.

Quantitative spatial relations. We connect non-hand physical entities with a spatial edge that carries a binary contact flag and a scale-ambiguous relative_vector $[\Delta x, \Delta y, \Delta z]$. The contact flag is computed in image space (mask touching or boundary distance below a small tolerance). To obtain relative_vector without intrinsics, human annotators associate each instance with a coarse 3D anchor in the single-image point set (Stage II); the vector is the anchor difference. This yields consistent near/far and directional cues sufficient for proximity-based feasibility checks (e.g., “pipette tip near tube rim”) without metric calibration or CAD priors.

Containment and partonomy. We add exactly one containment edge from each Substance to its Container, and exactly one part-of edge from each InteractiveElement to its parent object. Candidates are proposed by the VLM using node descriptions and mask geometry, then filtered by schema constraints (single parent; valid type roles). Ambiguities—typically due to occlusion among similar vessels—are resolved by human annotators.

Functional relations. Functional edges capture local interactions where the source is either an attached element (e.g., plunger to micropipette, cap to bottle, lid to dish) or a tool in immediate contact with its target during operation (e.g., pipette to tube when the tip is inside the opening). Candidates arise from spatial overlap or near-contact measured from masks and relative_vector. For each candidate, Gemini 2.5 Pro receives the cropped pair plus both node descriptions and returns (i) a feasibility flag consistent with the ontology and (ii) a short natural-language relation.description (e.g., “plunger controls liquid handling,” “pipette tip aspirates / dispenses inside the tube”). We keep edges that pass schema checks (valid roles; contact or immediate adjacency) and store the resulting relation.description on the edge to make the graph directly consumable by an LLM.

Physics-aware, state-conditioned affordances. We turn a descriptive graph into operational constraints for planning by defining three physics-aware gates as boolean validators computed only from fields already present in the graph (relative_vector,

contact, depth_avg, and, when applicable, fill_level). Each gate answers a task-level feasibility question with a yes/no decision. *Stability (Container).* A container c is stable if supported and not strongly overhanging in the image plane. Let $\mathcal{N}_{\text{sup}}(c)$ denote supporting objects with a spatial edge to c :

$$\text{Stability}(c) = \begin{cases} \exists s \in \mathcal{N}_{\text{sup}}(c) \text{ s.t. } \text{contact}(c, s) = 1 \\ \wedge \|\Pi_{xy}(\text{relative_vector}(c, s))\| < \tau_{\text{oh}}(\text{fill_level}(c)) \end{cases}$$

where Π_{xy} projects to the image plane and $\tau_{\text{oh}}(\cdot)$ optionally tightens with higher fill_level.

$$\text{Pourable}(c, t) = \begin{cases} \|\Pi_{xy}(\text{relative_vector}(t, c))\| < \tau_{\text{rim}} \\ \wedge (\text{contact}(t, c) \text{ or near-contact}(t, c)) \\ \wedge \text{fill_level}(c) < \tau_p \end{cases}$$

AspirateFeasible (Pipette, Container). A pipette p may safely aspirate from container c if the tip is near the opening and with non-trivial liquid:

$$\text{AspirateFeasible}(p, c) = \begin{cases} (\text{contact}(p, c) \text{ or near-contact}(p, c)) \\ \wedge \|\Pi_{xy}(\text{relative_vector}(p, c))\| < \tau_{\text{tip}} \\ \wedge \text{fill_level}(c) > \tau_a \end{cases}$$

Thresholds ($\tau_{\text{oh}}, \tau_{\text{rim}}, \tau_{\text{tip}}, \tau_p, \tau_a$) are tuned on a development split. Natural-language descriptions on nodes and edges (e.g., “can be capped for mixing,” “plunger controls liquid handling”) provide the *why*; the gates above provide the *can/can’t*. This binding converts descriptive structure into quantitative, planning-grade constraints that constrain plans, calibrate generative reasoning, and standardize evaluation—without expanding the JSON schema.

4.4 Stage IV: Quality Control and Tracks

Compilation track (knowledge base). Human annotators correct labels, refine fill_level for transparent liquids, and validate edges/affordances. A second-pass audit resolves disagreements; agreement statistics are reported with the knowledge base.

Serialization and artifacts. Each scene is exported as JSON file storing a P-3DSG and bundled with $\mathcal{I}$, the normalized depth D , and the generated point cloud $\tilde{\mathbf{X}}$. These artifacts constitute each WebLab-3DSG sample.

Complexity and runtime. On a single RTX 4090, Stage II detection+depth runs in $\sim 0.25\text{--}0.40$ s per image (batch size 1, FP16), relation verification adds $\sim 0.15\text{--}0.20$ s (depending on candidate pairs), and graph construction is < 50 ms. Confidence-aware re-prompts use $K \leq 3$ rounds for borderline pairs, adding < 200 ms overhead.

5 WebLab-3DSG

Scope and diversity. WebLab-3DSG comprises **1,000** single-view RGB scenes (one image per distinct lab setup) covering common ABL contexts (e.g., biosafety cabinets, open benches, object and liquid handlings, flame-based aseptic operations) with authentic hand-labware interactions. WebLab-3DSG will be released publicly.

Corpus-level statistics. Table 1 summarizes node/edge counts and distributions of HOI types and task categories. We also report aggregate liquid state and safety indicators. Table 2 highlights the most frequent entity labels and HOI tokens that appear in the knowledge base.

Scene-level example. A fully populated P-3DSG example, including node attributes and edge relations with both quantitative fields (e.g., `bbox_2d`, `depth_avg`, `rv`) and concise natural-language descriptions, is provided in Table 3.

6 Experiments

We evaluate whether grounding a language model in a P-3DSG improves web-image reasoning for laboratory manipulation. From **WebLab-3DSG**, we sample **100** distinct scenes and elicit **5** questions per scene authored by three senior lab experts (total **500** questions). The questions span five families designed to require quantitative attributes and state-conditioned affordances:

(1) **Liquid and containment attribution.** Identify the container–liquid pair and reason about amount using explicit containment and `fill_level`. Example: “Which vessel contains the red medium and what is its approximate fill level?”

(2) **Interactive elements and their functions.** Map elements to parent objects via `part-of/functional` edges and short relation descriptions. Example: “Which element controls the pipette and what is its role?”

(3) **State-conditioned feasibility.** Decide feasibility using the boolean gates `Stability`, `Pourable`, and `AspirateFeasible` that bind layout to liquid state. Example: “Is aspiration into the tube feasible with the pipette tip’s current position?”

(4) **HOI alignment and sterility practices.** Check whether hand posture, contact regions, and glove use align with sterile manipulation. Example: “Is the grasp posture appropriate for stabilizing the tube during aspiration?”

(5) **Safety constraints.** Assess support, overhang, and spill risk from the spatial graph. Example: “Does the current arrangement pose a spill risk if the container is tilted toward the pipette?”

Each family requires consulting graph attributes such as `containment`, `part-of`, `functional`, `relative_vector`, `contact`, and `fill_level`.

We compare two systems. **RGB+VLM** answers directly from the RGB image using **Qwen2.5-VL-7B-Instruct**. **P-3DSG+LLM** answers from the P-3DSG in JSON only using **Llama-3-70B-Instruct**. This pairing targets three goals: reproducibility, clean isolation of the graph signal (the graph system is text-only, avoiding hidden visual cues), and fairness in capacity. Frontier closed models (e.g., GPT-4o, Gemini) are not used in the main comparison to keep the protocol fully re-runnable.

Experts blind-rate each answer on a 0–5 rubric with three components: correctness (0–3; factual accuracy), justification (0–1; explicit citation of graph facts or visible evidence), and safety compliance (0–1; adherence to lab constraints). We report means across experts and scenes for each family and the macro-average across families.

Table 4 reports expert means (0–5; higher is better). P-3DSG+LLM outperforms RGB+VLM across all families by approximately **40%** on average. Gains are largest for state-conditioned feasibility, HOI alignment, and safety constraints, where the graph’s state variables

and boolean gates (`Stability`, `Pourable`, `AspirateFeasible`) provide auditable constraints. Improvements for liquid attribution and element function are smaller but consistent, reflecting the benefit of explicit containment and `part-of/functional` edges paired with short natural-language edge descriptions.

Table 4: Expert-rated QA on 100 scenes (500 questions). Mean score (0–5) $\uparrow$.

Question family	RGB+VLM	P-3DSG+LLM	Δ (%)
Liquid & containment	2.58	3.64	+41.1
Interactive elements & function	2.31	3.25	+40.7
State-conditioned feasibility	2.05	2.90	+41.5
HOI alignment & sterility	2.37	3.36	+41.8
Safety constraints	2.12	3.03	+43.0
Overall (macro-average)	2.29	3.24	+41.5

7 Conclusions and Limitations

We study a web-native question: how to turn single, uncalibrated RGB images from the open web into a queryable representation that supports physical reasoning. We introduce *WPIS*, a pipeline that compiles P-3DSGs from in-the-wild images. WPIS fuses open-vocabulary detection with monocular relative depth and a single-image 3D proxy to recover relative spatial layout, augments nodes with liquid *fill level* and fine-grained HOI subgraphs, and attaches concise natural-language functional relations with LLMs.

At scale, the same pipeline can support a community knowledge base. As more diverse scenes are compiled, the graph corpus becomes a shared, queryable reference for ABL planning and audit, and a source of structured priors for on-robot decision-making. To facilitate growth, we are launching a lightweight crowdsourcing portal where practitioners can upload single-view photos of real procedures with optional annotations. Submissions are screened for authenticity, anonymized, and aligned to the ontology so they can be merged into WebLab-3DSG and used by VLM/LLM agents in real ABL tasks.

The approach is not limited to biolabs. Any domain where fine manipulation depends on physical state—e.g., precision assembly or small-batch manufacturing—can adopt the same web-to-graph recipe: relative geometry, state variables, and short natural-language relations, plus domain-specific affordance gates. This yields a flexible knowledge base that strengthens physics-aware perception and planning without requiring instrumented capture.

Limitations remain. Single-image 3D perception in the wild is imperfect—occlusion, transparency, and highlights can bias quantitative node attributes. As transparent-object depth and open-vocabulary detectors improve, we expect fewer manual touches and tighter gates.

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Appendix

A Case study: a single biosafety-cabinet scene

Scene description and graph summary. Figure 4 shows a top-down view inside a biosafety cabinet. The P-3DSG for this scene contains Level-2 physical entities including a Container node for the *Angled-Neck Cell-Culture Flask* (T-flask), Tool nodes for the *motorized serological pipette controller* and the *held serological pipette*, two Hand nodes (left hand stabilizing the flask; right hand actuating the controller), and SupportingObject nodes for racks and reagent bottles. Level-3 nodes include a Substance node for *Red Cell-Culture Media* with *fill_level* = 0.1, and InteractiveElement nodes such as the *serological pipette* (attached to the controller) and the *flask cap* (removed and placed on the surface). Functional and part-of edges bind the controller to the pipette and the cap to the flask; spatial edges capture tip–mouth proximity.

Comparison. On this scene, RGB+VLM (Qwen2.5-VL-7B-Instruct) achieved an expert mean of **1.90/5**. It correctly attributed the liquid to the T-flask and inferred a sub-50% fill level, but misclassified the right-hand tool as non-motorized, judged aspiration feasible without adjustment, and expressed uncertainty about tilt stability. P-3DSG+LLM (Llama-3-70B-Instruct) achieved an expert mean of **4.24/5**. It leveraged explicit graph fields—*content_1.fill_level* = 0.1, *part-of/functional* links between controller and pipette, and the pipette–flask *relative_vector*—to recognize the motorized controller, judge the tip as inside/near-contact, and recommend lowering the tip before aspiration to avoid air intake. Both systems recommended placing the cap face-down for sterility; the remaining ambiguity is that the graph does not encode cap orientation, which limits perfect justification on that item.

This appendix presents the complete QA artifact for that single biosafety-cabinet scene: the five expert-authored questions (one per family), the expected answer formats, the verbatim outputs from **RGB+VLM** (Qwen2.5-VL-7B-Instruct) and **P-3DSG+LLM** (Llama-3-70B-Instruct), and the per-expert 0–5 scores together with their mean. Wording is reproduced exactly as generated by each system; no post-editing is applied. See Table 5 for the full listing.



Figure 4: Scene used in the case study. A motorized serological pipette controller approaches an angled-neck T-flask containing red medium on the work surface; tube racks and reagent bottles are in the background.

B Interactive 3D Visualization

Since the three-level schema maps cleanly to a spatial layout, we developed a browser-based interactive 3D viewer in Figure 5. This tool renders different types of nodes and edges, allowing users to filter attributes, toggle edge types (e.g., showing only functional links), and inspect the spatial consistency of the graph. This viewer will be released alongside the WebLab-3DSG dataset.

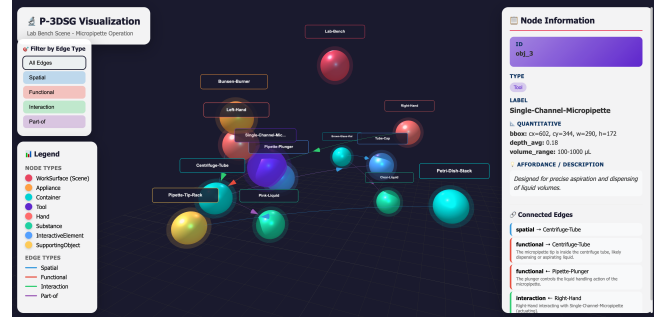


Figure 5: The 3D visualization tool showing the interactive visualization of the reported example P-3DSG in Table 3.

C Extended Implementation Details

C.1 Depth Estimation Modules

Our pipeline adopts a modular design for relative depth estimation. By default, we employ MODEST [9] (trained on ClearPose) because its domain adaptation specifically handles transparent labware (e.g., glass tubes, plastic bottles) common in web-sourced lab images. However, the module is swappable: for broader, non-transparent environments, Depth_Anything_v2 [59] serves as a strong alternative due to its large-scale open-world pre-training. The P-3DSG schema is modality-agnostic; while the current pipeline relies on single-view RGB, it can optionally ingest metric depth to override relative vectors when RGB-D sensors are available, increasing fidelity without altering the graph representation.

C.2 Automatic Fill Level Estimation

While the WebLab-3DSG knowledge base is curated with human verification to ensure ground-truth quality, the inference pipeline operates automatically. At deployment, the liquid *fill_level* attribute is estimated using a segmentation-based approach, calculating the ratio of the liquid mask area to the container mask area. This estimate is further stabilized using relative-depth ordering to handle occlusion.

D Human Evaluation and Data Curation

D.1 Expert Rater Qualifications

The human evaluation reported in Section 6 was conducted by three biomedical engineering PhDs. Each expert possesses 8–12 years of wet-lab experience spanning both academic research and industrial pharmaceutical settings. Prior to scoring, raters were calibrated on a held-out set of examples to resolve disagreements and finalize the grading rubric.

Table 5: Scene QA — expected formats and verbatim answers from Qwen2.5-VL-7B-Instruct (RGB+VLM) and Llama-3-70B-Instruct (LLM+P-3DSG), with expert mean scores (0–5) shown next to each answer.

#	Family	Question	Expected format	Qwen2.5-VL-7B-Instruct (RGB only)	Llama-3-70B-Instruct (+P-3DSG)
1	Liquid & containment	Which object contains the pink/red liquid, and is its fill level < 50% or ≥ 50%?	<object_label>, <less/greater-than-50%> + short justification.	Answer: T-flask, < 50%. The liquid level in the T-flask appears to be below the midpoint, indicating less than 50% fill level. (score: 2.40)	Answer: Angled-Neck-Cell-Culture-Flask, less than 50%. The pink liquid is contained in 'obj_1' (the flask); 'content_1' has 'fill_level: 0.1' (Red-Cell-Culture-Media). (score: 4.60)
2	Interactive elements & function	Is the right-hand tool a motorized serological pipette controller attached to a serological pipette? And is the pipette tip inside/near the mouth of the T-flask?	yes/no + inside/near/outside + one-sentence rationale.	Answer: no (reagent bottle), away. The T-flask is used as a container for aspirating liquid, not a motorized serological pipette controller. The tool is placed below the T-flask opening on the work surface, suggesting no motorization is attached. (score: 1.30)	Answer: yes, inside/near-contact. The right-hand tool is a motorized serological pipette controller ('obj_3') attached to a serological pipette ('elem_1'); tip near/inside the T-flask mouth ('obj_1') supported by the relative vector '[0.429, 0.102, 0.090]'. (score: 4.20)
3	State-conditioned feasibility	Given the current arrangement, is aspiration without air intake feasible immediately, or should the tip be lowered/angle adjusted first?	yes/no + one actionable adjustment, if any.	Answer: yes, no action required. The tip is well inside the T-flask, and the flask is level, so aspiration should be feasible without further adjustments. (score: 1.60)	Answer: no, lower tip. The flask is angled and the tip has not yet fully contacted the liquid surface ('fill_level=0.1'); insert slightly deeper; avoid further tilting. (score: 4.30)
4	HOI alignment & sterility practice	Is the T-flask cap placed in a sterility-consistent way (e.g., face down on a clean surface)?	yes/no + "face-down recommended"/"acceptable as is" + rationale.	Answer: no, face-down recommended. If the cap is off on the surface, it should be face down to avoid contamination. Current face-up placement poses a risk. (score: 2.30)	Answer: yes, face-down recommended, to prevent contamination. The cap ('elem_3') is off and resting on the surface; in a BSC it should be face down. Not specified in graph whether current placement is acceptable. (score: 4.00)
5	Safety constraint (tilt stability & spill risk)	If the operator tilts the T-flask by ~ 30° toward the pipette, is the setup stable and spill-safe given the current fill level and support?	yes/no + brief reason.	Answer: no; lower the fill below headspace. Tilting by 30° might be unstable if there's too much headspace liquid; ensure level below headspace to reduce spill risk. (score: 1.90)	Answer: yes. With 'fill_level=0.1' and a supporting hand ('obj_4') in contact, stability and spill safety are maintained when tilting toward the pipette. (score: 4.10)

D.2 Schema Extensibility

Our internal audit revealed that certain specialized questions may depend on attributes beyond the core set. The P-3DSG schema supports optional, modular fields. Identified candidates for future inclusion are: *cell culture type* (adherent vs. suspension), *container tilt angle*, *material transparency*, *pipette volume setting*, and the

presence of *foam*. These fields can be instantiated when detected by task-specific VLM prompts.

E Reproducibility

We will release Images, graph JSONs, prompts, scoring rubrics, scene IDs, and code to format P-3DSG JSON into the LLM prompt. Random seeds and model checkpoints are fixed for both systems.